

What the prototype must prove

First milestone: one safely assembled indoor unit with recorded airflow, filter pressure loss, electrical power, temperature and repeatable particle trends. Not a full outdoor tower and not a certified HEPA air cleaner.

This delivery: current R03M pressure screen, blank measurement workbook (three CSV sheets), a strict airflow calculator, synthetic regression tests and one physical test sequence. No fabricated results fill the empty fields.

Important correction: both guard patterns now use 0.0354623 m² open area each, not the earlier 0.036 m². With the same assumed loss coefficient, each guard loss increases by about 3.06%. This does not verify the loss coefficient or actual guard resistance.

Our project: a custom cabinet, seals, service arrangement, guarded four-fan assembly and measured system integration. Buying OEM fans/filters is normal; it does not make the system a proven product or a patentable invention.

Stop condition: do not energize the review-only assembly to collect these data. Mechanical strength/guard/seal review and the rated protective electrical design remain open. The test workbook is ready; the hardware is not yet released.

Evidence labels: OEM = published component data; ASSUMED = modeling choice; CALCULATED = arithmetic from stated inputs; MEASURED = physical record with method/instrument evidence. At present, physical prototypes and physical test runs both equal zero.

Updated pressure budget

Calculated, not measured: four fans in parallel add flow, not pressure. Two filter paths each carry half the total flow only under the equal-sharing assumption. Filter pressures must not be added as if the two paths were in series.

Central assumed reducer K = 1:

Total 200 m³/h / each filter 100: fan 32.60 Pa; other losses 5.92 Pa; filter allowance **26.68 Pa**.

Total 250 m³/h / each filter 125: fan 30.09 Pa; other losses 9.24 Pa; filter allowance **20.85 Pa**.

Total 300 m³/h / each filter 150: fan 27.63 Pa; other losses 13.31 Pa; filter allowance **14.32 Pa**.

Total 350 m³/h / each filter 175: fan 25.27 Pa; other losses 18.12 Pa; filter allowance **7.15 Pa**.

Total 400 m³/h / each filter 200: fan 22.83 Pa; other losses 23.66 Pa; filter allowance **-0.83 Pa**.

Meaning: the filter must fit inside the remaining pressure allowance at that same flow in this model. Negative allowance means the assumed non-filter losses already exceed the curve pressure. Positive allowance is NOT proof that the actual filter fits.

Model assumptions: OEM numeric curve at full 2800 rpm; density 1.2 kg/m³; 134.2 mm fan apertures; 350 x 270 mm opening per filter. Guard K1.5 each, plenum K1.5 and installation K1 on their stated velocity references. Reducer K0.5/1/2 sensitivities are included in the CSV.

Still unknown: actual installed fan curve, clean/loaded filter curve, guard K, bypass, uneven flow sharing and outlet recirculation. Reducer and plenum losses may overlap. Static pressure compatibility remains unresolved. These are not guaranteed bounds or a new CFD validation.

Decision: keep 300 m³/h as a provisional investigation point, not a product claim. Do not increase the power supply or fan speed beyond OEM limits to force that result. Change the design if actual flow/pressure evidence demands it.

Measure flow before claiming cleaning

1. Identify the exact assembly. Record CAD revision, delivered fan/filter identities, seals, guards, speed setting, room conditions and instrument/calibration references. Photograph the complete guarded configuration. Keep raw records unchanged.

2. Use a reviewed flow method. Prefer a suitable calibrated capture arrangement or an engineered duct traverse. A handheld reading at one outlet point multiplied by gross area is not a valid total flow measurement. Swirl, nonuniform velocity and both guard layers matter.

Fixture warning: a hood or duct can restrict this low-pressure fan system. Record fixture resistance and use a method validated for the installed configuration; do not call a loaded-fixture flow the unrestricted device flow without justification. Fixture dimensions and correction are not yet selected.

3. Record area-weighted cells. AIRFLOW_BLANK.csv contains empty placeholders, not an approved nine-point traverse. For an accepted normal-velocity map, $Q = 3600 \times \sum(\text{cell area} \times \text{mean normal velocity})$. Cells must cover the chosen plane once without overlap; the reviewer determines sampling and averaging.

4. Quantify error. The calculator reports only a conservative velocity-error contribution, $3600 \times \sum(\text{area} \times \text{velocity error bound})$. Area, temporal drift, probe alignment, fixture loading/leakage and spatial sampling require a separate full uncertainty budget. It does not manufacture a confidence interval.

5. Record pressure and power. Measure each filter-path pressure drop separately using reviewed static tap positions. Do not drill before sealing/reach review. Record total electrical input with a suitable rated instrument; 16.8 W nominal fan load is not measured wall power.

Particle trial: useful, not certification

Prepare a controlled room. Record room dimensions, sensor positions, ventilation/HVAC, doors/windows and occupancy. Compare sensors together beforehand; document their response, limitations and detection floor. Keep locations and room conditions comparable across runs.

Do not generate smoke or dust yourself. No incense, combustion, powders or deliberate pollutant release in an occupied room. A formal challenge-aerosol test belongs with an equipped competent service. If ambient particles are too low or unstable, mark the trial inconclusive.

Suggested engineering trial: paired device-OFF baseline and device-ON observation windows, repeated under comparable conditions, using naturally present particles. Log raw PM, temperature and humidity; do not choose only the best-looking interval. Duration must provide adequate signal while conditions remain stable; no invented universal duration.

Analysis boundary: a PM decrease may reflect settling, ventilation, changed sources or sensor humidity response. A simple before/after percentage is not device efficiency. Blank particle data are supplied, but no automatic CADR calculator is issued before the room method and uncertainty are defined.

What can be reported: dated observations, run conditions, measured flow/power, particle trends and limitations. Formal CADR needs a validated method; HEPA-level efficiency needs appropriate filter/installed-leak evidence. Airflow alone proves neither.

No outdoor bubble claim: indoor results cannot establish a public-space radius or tower spacing. Heat, weather, wind, public access, stability and maintenance are separate design and test problems. Carbon/gas removal is outside this particulate trial.

Acceptance, tools and handoff

Before testing: rated protective design and qualified commissioning; guard/access/strength/seal closure; calibrated instruments and a reviewed test method. Funding-only support means rent instruments or commission an equipped service; no college lab is assumed.

Instrument functions, not purchase orders: total-flow measurement suitable for low-pressure fans; differential pressure measurement resolving expected small losses; rated input-power measurement; temperature/humidity logging; comparable particle monitors; noise meter if reporting noise. Exact ranges/accuracy and availability must match the approved method.

Acceptance rules: safety faults mean STOP. Missing method, invalid calibration, leaks, sensor saturation, unstable room conditions or insufficient particle signal mean INVALID/INCONCLUSIVE, not PASS. Flow below a project target requires design review; the 300 m3/h investigation point is not a safety or certification threshold.

Reproducible tools: run scripts/analysis/d01_validation.py for synthetic tests. With accepted physical data: add --airflow-csv FILE --output NEW_FILE.json. Existing output files are not overwritten. Blank templates, NaN, missing evidence, duplicate cells and mixed runs are rejected.

Checks completed: 15 synthetic/arithmetic checks. Input hashes link the pressure screen to the OEM CSV and current guard pattern. No physical measurements or tests performed; no purchases, installation or supplier messages.

Three packages now exist, not three approvals: mechanical R03M; electrical E00; validation V00. Remaining critical path is actual protective/structural/interface closure, qualified review, authorized build, commissioning and measured performance. More PDFs cannot substitute for those steps.

Sources: archived ARCTIC numeric workbook (source URL/hash in JSON); R03M guard_pattern.json; [EPA Guide to Air Cleaners](#) and [EPA DIY Air Cleaners](#), checked 4 October 2026. EPA guidance is not approval of D01. Test procedures here are proposed project methods, not copied certification standards.